Journal of Ethnopharmacology 263 (2020) 113158

Contents lists available at ScienceDirect

Journal of Ethnopharmacology

journal homepage: www.elsevier.com/locate/jethpharm

Danlou tablet inhibits the inflammatory reaction of high-fat diet-induced atherosclerosis in ApoE knockout mice with myocardial ischemia via the NF-κB signaling pathway

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ARTICLE INFO

Keywords:
Danlou tablet
Atherosclerosis
Myocardial ischemia
ApoE
High-fat diet
NF-κB signaling pathway

ABSTRACT

Ethnopharmacological relevance: Danlou tablet (DLT), a traditional herbal formula, has been used to treat chest discomfort (coronary atherosclerosis) in China. Although the anti-inflammatory activities of DLT have been proposed previously, the mechanisms of DLT in treating atherosclerosis with myocardial ischemia (AWMI) remain unknown.

Aim of the study: Atherosclerosis can result in heart disease caused by stenosis or occlusion of the lumen, resulting in myocardial ischemia, hypoxia, or necrosis. In recent years, changes in people's diets, increased stress, and secondary fatigue and obesity etc. have resulted in increases in the number of patients with atherosclerosis. In cases where the condition has further developed, patients may suffer from myocardial ischemia, hypoxia, or necrosis. Many traditional Chinese medicine compounds have been prescribed for the treatment of AWMI. DLT has been used to treat chest discomfort (coronary atherosclerosis) in China. Based on previous research, the aim of this study was to further investigate the effect of DLT on AWMI, and describe the underlying mechanisms.

Materials and methods: To achieve this, an animal model of AWMI was established using apolipoprotein E ($ApoE^{-/-}$) mice fed a high fat diet combined with isoprenaline (ISO) injection. For comparison, mouse models of only atherosclerosis and only myocardial ischemia were included. In the treatment groups, mice were treated daily with DLT at 700 mg/kg for four weeks. Echocardiographic evaluation, hematoxylin and eosin (H&E) staining, oil red O staining, ELISAs, Western blots, and immunohistochemical analyses were subsequently used to investigate the mechanism of DLT based on the NF-κB signaling pathway.

Results: The results indicate that the use of DLT is effective, to varying degrees, for the treatment of atherosclerosis, myocardial ischemia, and AWMI in mice. After DLT treatment, the left ventricular structure and morphology of the mice, the histopathology of cardiac tissue, and atherosclerotic plaques in the aortas all improved to varying degrees. DLT could play a therapeutic role by regulating the NF-κB signaling pathway related to inflammatory factors, including TNF-α, IL-6, IL-1β, IL-8, MMP-1 and MMP-2, as well as protein

Abbreviations: AWMI, atherosclerosis with myocardial ischemia; DLT, Danlou tablet; $ApoE^{-/-}$, apolipoprotein E; ISO, isoprenaline; H&E, hematoxylin and eosin; TC, total cholesterol; TG, triglyceride; LDL-C, low density lipoprotein-cholesterol; TNF, tumor necrosis factor; IL, Interleukin; MMP, matrix metallopeptidase; LVM, left ventricular masses; LV Vold, left ventricular end-diastolic volumes; LV Vols, left ventricular end-systolic volumes; LVIDd, left ventricular internal dimensions at end-diastole; MMPs, matrix metalloproteinases.

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https://doi.org/10.1016/j.jep.2020.113158
Received 22 December 2019; Received in revised form 8 June 2020; Accepted 2 July 2020
Available online 31 July 2020

expression of NF-κB p-50 and IκB-α, and positive cell expression of NF-κB p-50, IκB-α and phospho-NF-κB p-50 in the model mice.

Conclusion: These preliminary results indicate that the therapeutic efficacy of DLT on high-fat diet-induced atherosclerosis in ApoE$^{-/-}$ mice with myocardial ischemia could be exerted at least in part by regulating the NF-κB signaling pathway.

1. Introduction

The 2016 Global Burden of Disease Study shows that, between 2006 and 2016, the number of deaths from cardiovascular diseases increased by 14.5% globally, and cerebrovascular diseases (17.6 million) are the chronic non-communicable diseases that cause the highest number of deaths. Ischemic heart disease and cerebrovascular disease (stroke) together accounted for 85.1% of all cardiovascular disease deaths in 2016 (GBD, 2016 Causes of Death Collaborators, 2017). Cardiovascular disease is the main cause of death in today's society (Benjamin et al., 2019; Huang et al., 2019). Its main pathological basis is coronary atherosclerosis. Modern medicine believes that coronary heart disease not only has abnormal hemodynamics, but also has obstacles such as immunity, endocrine, and metabolism of multiple substances. In addition, pathological studies and epidemiological investigations show that inflammation plays an important role in the development of atherosclerosis, and atherosclerosis in patients is usually accompanied by local and systemic inflammatory reactions (Ross, 1999; Moore et al., 2013; Kwon et al., 2017). Inflammatory factors, as an indicator of inflammatory response, are a class of endogenous polypeptides which are mainly produced by immune cells and have strong biological effects. They can mediate multiple immune responses, and are closely related to the occurrence and development of cardiovascular diseases (Shirai, 2015; Li et al., 2017). Therefore, it is important to explore the relationship between various inflammatory factors and myocardial ischemia in patients with atherosclerosis to achieve early diagnosis, treatment, and improve prognosis.

Moreover, it has been firmly established that inflammation plays an important role in the development of atherosclerosis, and various inflammatory factors can accelerate the development of atherosclerosis, leading to cardiovascular adverse events. NF-κB is an important inflammatory transcription factor, and it is a dimer composed of polypeptide chains p-65 and p-50 protein subunits. When NF-κB is stimulated by cytokines such as tumor necrosis factor (TNF)-α and interleukin (IL)-1, it shifts into the nucleus and regulate the expression of various inflammatory factors, adhesion molecules, and matrix metalloproteinases (MMPs) at the transcription level, so NF-κB is one of the key targets in the mechanism of vascular endothelial damage (García de Tena, 2005; Feng et al., 2017; Bhat et al., 2018). NF-κB and the inflammatory factors expressed by its target genes interact to further promote the progression of atherosclerosis (Tak and Firestein, 2001; Li et al., 2004). Through a review of the relevant literature on the NF-κB signal transduction pathway, it was found that the NF-κB signal transduction pathway is involved in the occurrence and development of different forms of coronary heart disease. Therefore, it can be speculated that the NF-κB signal transduction pathway may be closely related to the occurrence of coronary heart disease with atherosclerosis with myocardial ischemia (AWMI), and NF-κB is an important trigger point in the NF-κB signal transduction pathway and participates in the overall regulation of this pathway. Therefore, we propose the following hypothesis: NF-κB is a new target for the modulation and treatment of AWMI. Intervention with NF-κB signaling pathway may be a new strategy for the treatment of AWMI.

Danlou tablet (DLT) originated from a famous ancient traditional Chinese medicine prescription, which is the Gualou-Xiebai-Banxia decoction. This decoction is described in the *Synopsis of Prescriptions of the Golden Chamber* (Jin-Gui-Yao-Lue) written by Zhongjing Zhang, a Chinese medical scientist in ancient China, and these treatments have been used for approximately 2000 years (Mao et al., 2016; Wang et al., 2016). DLT has been widely sold to treat cardiovascular disease, such as coronary heart disease, myocardial ischemia, atherosclerosis, angina pectoris, and hyperlipidemia in many hospitals and drugstores (Wang et al., 2015a,b,c). It is approved by the China Food and Drug Administration (No. Z20050244) and is composed of ten herbs, including *Trichosanthes kirilowii* Maxim. (Cucurbitaceae), *Allium macrostemon* Bunge. (Liliaceae), *Pueraria lobata* (Willd.) Ohwi (Leguminosae), *Ligusticum chuanxiong* Hort. (Umbelliferae), *Salvia miltiorrhiza* Bunge. (Labiatae), *Paeonia lactiflora* Pall. (Ranunculaceae), *Alisma plantago-aquatica* Linn. (Alismataceae), *Astragalus membranaceus* (Fisch.) Bunge. (Leguminosae), *Davallia mariesii* Moore ex Bak. (Davalliaceae), and *Curcuma aromatica* Salisb. (Zingiberaceae). The high performance liquid chromatography (HPLC), ultra high performance liquid chromatography/diode-array detector/quadrupole time-of flight tandem mass spectrometry (UPLC-DAD-QTOF) and gas chromatography-mass spectrometry technology (GC-MS) were used to identify the major constituents of DLT for the quality control (Dong et al., 2013; Gao et al., 2015; Zhang et al., 2016a,b). Previous reports indicated that main chemical constituents in DLT include gallic acid, danshensu, 5-hydroxymethyl-2-furaldehyde, 3′-hydroxy puerarin, puerarin, 3′-methoxy puerarin, mirificin, daidzin, paeoniflorin, calycosin-7-*O*- β-D-glucoside, naringin, genistin, rosmarinic acid, salvianolic acid B, salvianolic acid A, formononetin, calycosin, cryptotanshinone and tanshinone IIA, and among them, the component with the highest content in DLT is puerarin (Dong et al., 2013; Gao et al., 2015; Li et al., 2019). Therefore, the puerarin as the major component of DLT was analyzed and determined for quality control.

Previously, we designed inflammatory models induced by LPS *in vivo* and *in vitro*, and found that Danlou prescription ethanol extract (EEDL) had anti-inflammatory effects, and the potential molecular mechanisms might be due to the inhibition of LPS-induced iNOS/NO, COX-2/PGE_2 and cytokines expression by antagonizing the activation of NF-κB p65 (Gao et al., 2015). Furthermore, we found that EEDL could inhibit foam cell formation induced by ox-LDL via the TLR4/NF-κB and PPARγ signaling pathways (Gao et al., 2018). Although the anti-inflammatory effect has preliminarily been revealed as one of the potential mechanisms of the treatment effects of DLT on coronary heart diseases, the underlying mechanisms of the treatment of AWMI by DLT remain unclear. Therefore, in this study, apolipoprotein E knockout (ApoE$^{-/-}$) mice were fed high fat feed and injected with isoprenaline (ISO), were treated with DLT. Furthermore, we investigated the effects and mechanisms of DLT treatment on AWMI by echocardiographic evaluation of cardiac function, observing the histopathology of the heart as well as atherosclerotic plaques in the aorta, and detecting the inflammatory factors regulated by the NF-κB signaling pathway.

2. Materials and methods

2.1. Drugs and reagents

DLT (batch number: 20140302) was provided by Jilin Connell Me dicine Co., Ltd. (Jilin, China). For quality control, the major component of DLT was analyzed and determined by ultra-performance liquid chromatography quadrupole time-of-flight mass spectrometry (UPLC-Q-TOF/MS) (Dong et al., 2013; Gao et al., 2015; Li et al., 2019). DLT powder was suspended and diluted with physiological saline, using an ultrasonic water bath for administration to mice. ISO was purchased from Sigma-Aldrich Co. (batch number: I5627-5G). Atherosclerotic molding feed was purchased from Jiangsu Medisen Biomedical Co., Ltd. (21% milk fat, 0.15% cholesterol, 20% protein, 50% carbohydrate, 4.7 kcal/mg, batch number: MD12015). A hematoxylin and eosin (H&E) staining kit was purchased from Beyotime Biotechnology (Shanghai, China). Oil red O was obtained from Sigma–Aldrich Co. (St. Louis, MO, USA). ELISA kits for total cholesterol (TC), triglyceride (TG), and low density lipoprotein-cholesterol (LDL-C) were purchased from Biosino Bio-Technology and Science Inc. (Beijing, China). Mouse Platinum ELISA kits for TNF-α, IL-6, and IL-1β were purchased from eBioscience Co. (CA, USA). Mouse ELISA kits for IL-8 were purchased from Neo-bioscience Co. Ltd. (Shenzhen, China). Mouse ELISA kits for matrix metallopeptidase (MMP)-1 and MMP-2 were purchased from Cloud-Clone Coup. (Wuhan, China). A BCA Protein Assay Kit was purchased from Thermo Fisher (Rochford, USA). The antibodies for NF-κB p-50 (catalogue number: ab32360) and IκB-α (catalogue number: ab32518) were purchased from Abcam (Cambridge, UK).

2.2. Mice

Six-week old male C57BL/6J mice and six-week old male ApoE$^{-/-}$ mice (weighing 18–22 g) were purchased from Beijing Vital River Laboratory Animal Technology Co., Ltd. (Beijing, China, Certificate No. SCXK Jing, 2012-0001). All the mice were housed in cages at a temperature of 24 ± 1 °C with 55% ± 5% humidity, under a 12-h light/dark cycle. Food and water were freely available, except during testing. The mice were acclimatized to the laboratories for one week before the experiment. Studies were performed in accordance with NIH Guide for the Care and Use of Laboratory Animals, and protocols were approved by the Ethics Committee of Tianjin University of Traditional Chinese Medicine (TCM-LAEC20170053).

Mice were divided into seven groups, including 1) Control group (15 C57BL/6J mice): abbreviated as CG, normal feed; 2) High fat feed group (14 ApoE$^{-/-}$ mice): abbreviated as HG, atherosclerotic molding feed; 3) ISO group (13 ApoE$^{-/-}$ mice): abbreviated as IG, intradermal injection of ISO at a dose of 10 mg/kg per day for six consecutive weeks (once daily, note that an equal amount of 0.9% saline was administered to mice in the CG and HG groups); 4) High fat feed + ISO group (14 ApoE$^{-/-}$ mice): abbreviated as HIG, atherosclerotic molding feed and ISO administered at a dose of 10 mg/kg; 5) High fat feed + DLT group (14 ApoE$^{-/-}$ mice): abbreviated as HDG, atherosclerotic molding feed and DLT administered daily at 700 mg/kg (via intragastric gavage, for 7–10 weeks. Note that an equal amount of 0.9% saline was administered to mice in the CG, HG, IG, and HIG groups); 6) ISO + DLT group (13 ApoE$^{-/-}$ mice): abbreviated as IDG, ISO administered daily at 10 mg/kg (note that an equal amount of 0.9% saline was administered daily to mice in the HDG group) and DLT administered daily at 700 mg/kg; 7) High fat feed + ISO + DLT group (14 ApoE$^{-/-}$ mice): abbreviated as HIDG, atherosclerotic molding feed, ISO administered daily at 10 mg/kg, and DLT administered daily at 700 mg/kg (Zhang et al., 2016a,b). In this study, the HG, IG and HIG groups were classified as model groups, and the HDG, IDG and HIDG groups were classified as treatment groups.

One hour after the last drug administration, blood samples were collected from mice. The mice were executed by rapid dislocation of the cervical spine after taking blood. Serum was obtained by incubating blood samples for 1 h at 37 °C (water bath), and then centrifuging samples at 3500 × g for 10 min at 4 °C. Serum samples were stored at −20 °C until use. Simultaneously, some of the cardiac tissues and aortas of the mice were quickly removed, immediately frozen in liquid nitrogen, and stored at −80 °C until analysis. The other parts of the cardiac tissues and aortas were fixed with 10% formalin solution for H&E staining or oil red O staining. A schematic diagram of the study is shown in Fig. 1.

2.3. Detection methods of serum lipids in mice

The levels of TG, TC and LDL-C in the serum of mice were detected with the kit on day 42 of the experiment. The samples were processed according to the kit instructions and the OD values were measured using a multi-well microplate analyzer, and the content of TG, TC and LDL-C in the samples was calculated according to the standard curve.

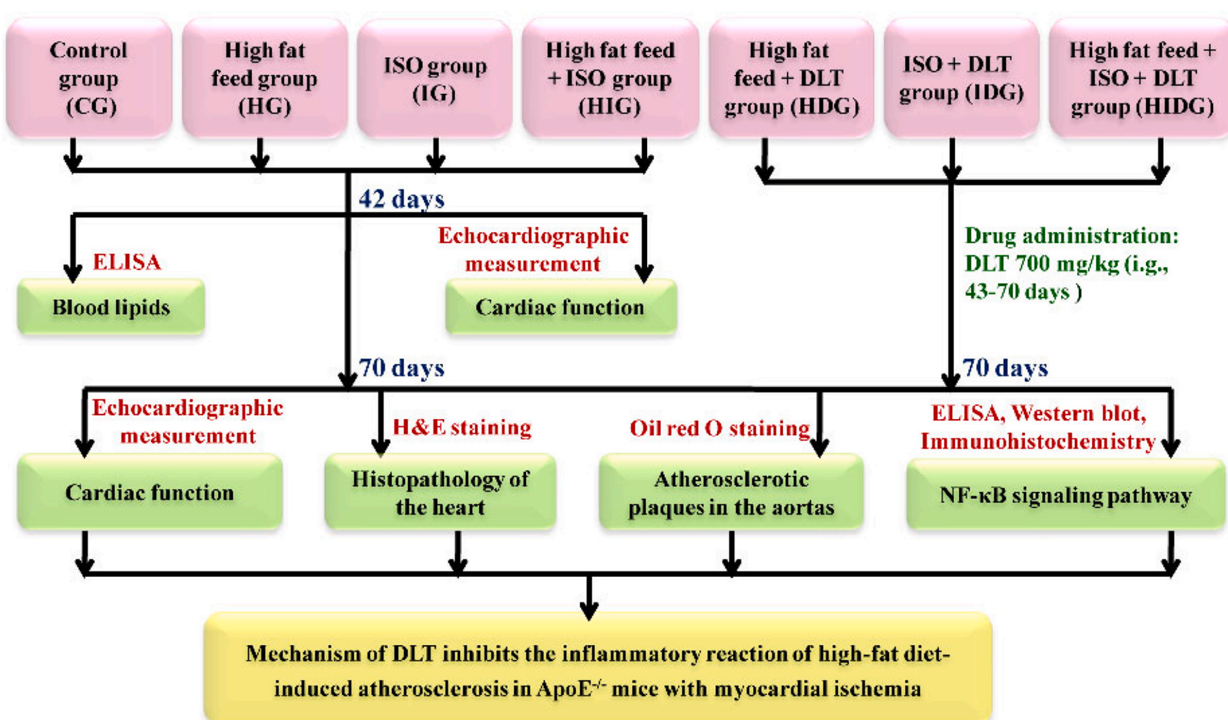

Fig. 1. Schematic diagram of the study.

2.4. Echocardiographic measurement

Fifteen minutes after subcutaneous injection of ISO on day 42 and 70 of the experiment, respectively, the mice were anaesthetized with tribromoethanol in order to evaluate their heart function using an ultra-high resolution small animal ultrasound imaging system (Vevo2100, Visual Sonics, Canada). The chest hair of mice was shaved with depilation cream, and the chest area was cleansed. The mice were kept at a constant temperature (maintained at 37 °C), and physiological parameters such as electrocardiogram and respiration were recorded simultaneously. The heart rate was maintained at 300 to 400 beats/min. After the heart rate was stable for 1 min, a uniform coating of medical ultrasonic coupling agent was applied to the chest for ultrasound examination (Zhang et al., 2016a,b).

The examination table was placed horizontally. The probe was parallel to the long axis of the mouse's body, and placed on the mouse's thoracic shank. The probe was at an angle of 30° to 45° from the mouse's left chest wall, and the X and Y direction knobs were adjusted. The long axis section of the heart was obtained under B-Mode. The papillary muscle was used as an anatomical mark to facilitate the accurate location of the measurement site in the long axis of the heart. Dynamic images were collected from all the sections continuously for 10 s. After storing the images, an image analysis system (Visualsonics, Toronto, Canada) was used for offline analysis (Hu et al., 2016; Yu et al., 2017).

2.5. Histopathological examination

On day 70 of the experiments, the cardiac tissues of mice were preserved in 10% formalin for H&E staining, and tissue sections with a thickness of 2 to 3 mm were cut. We placed tissue sections in 75% ethanol for 1 h, followed by overnight incubation in 95% ethanol, incubation in ethanol I for 1 h, incubation in ethanol II for 4 h, incubation in xylene I for 15 min, incubation in xylene II for 45 min, incubation in soft wax for 30 min, and finally incubation in hard wax for 1 h. Finally, we poured melted paraffin into the bury frame, and then immediately placed the hard-waxed tissue block into the melted paraffin. After the wax block solidified, it was fixed on a paraffin slicer, and tissue slices were cut at a thickness of 5 μm. Tissue slices were separated in warm water (49 °C). Tissue sections were roasted for 60 min in an electrothermal drying oven (60 °C), and subsequently stored at room temperature after cooling. We placed the paraffin sections in an oven at 60 °C for 30 min, soaked them in dimethyl benzene solution I, II, dewaxed the sections for 10 min, and then incubated sections in 100% ethanol I, 100% ethanol II, 95% ethanol, 75% ethanol gradients for 5 min, after which we rinsed the sections for 3 min with running water. The paraffin sections were soaked in the hematoxylin dyeing solution for 8 min, followed by differentiation in 1% ethanol hydrochloride for 15 s, rinsing with running water for 20 min, staining in 1% eosin solution for 1 min, rinsing with running water for 3 min, uplink gradient dehydration in 75% ethanol for 2 s, 95% ethanol for 2 s, and 100% ethanol I, II for 2 min, and xylene I and II were applied 5 min apart to make the sections transparent. Neutral gum was used to seal the tissue sections, and sections were observed and photographed under a light mirror (Ma et al., 2016; Du et al., 2018; Liu and Qin, 2018).

2.6. Oil red O staining

On day 70 of the experiment, the entire aorta was quickly removed, and the remaining tissue around the aorta was dismembered. The aorta was washed with cold saline solution and preserved in 10% formalin for oil red O staining. The ratio used to make a working dilution of the red O dye was 3:2 (red O dye:distilled water). The ratio of isopropanol to distilled water was 3:2, which formed a 60% isopropanol solution. Subsequently, the aorta was removed from the 10% formalin solution, rinsed for 1 to 2 min with distilled water, dipped in 60% isopropanol for 1 min, stained with the oil red O working dilution for 10 min, and then the aorta was placed in 60% isopropanol. Aortas were differentiated until the vascular wall of the aortas was transparent, and the difference between the color and the plaque was clear. Finally, the aortas were stored in 10% formalin, after staining and taking photos. Simultaneously, using Image-Pro Plus Version 6.0 software, the red plaque area was marked, and the ratio of plaque area to vascular wall area was calculated.

2.7. ELISA for TC, TG, LDL-C, TNF-α, IL-6, IL-1β, IL-8, MMP-1 and MMP-2

On day 42 of the experiment, Mouse ELISA kits were used to detect TC, TG, and LDL-C concentrations in serum of mice, according to the manufacturer's instructions. On day 70 of the experiment, the TNF-α, IL-6, and IL-1β concentrations in cardiac tissues of mice were assayed using the Mouse Platinum ELISA kit, according to the manufacturer's instructions. The Mouse ELISA kit was used to detect IL-8 concentrations in serum of mice, according to the manufacturer's instructions. Concentrations of MMP-1 and MMP-2 in serum of mice were analyzed using Mouse ELISA kits, according to the manufacturer's instructions.

2.8. Western Blot analyses

On day 70 of the experiment, cytoplasmic protein of the cardiac tissue was precipitated with isopropyl alcohol for 10 min, followed by centrifugation at 12,000 × g at 4 °C for 10 min. Next, guanidine hydrochloride (0.3 M) was added to denature the protein for 20 min. Protein precipitation was achieved by centrifugation at 7500 × g for 5 min. Thereafter, the protein precipitate was washed with ethanol. After aspirating the ethanol, 1% SDS was added to dissolve the protein at 50 °C. The protein concentration was determined using a BCA assay (Pierce, United States). Equal amounts (20 mg) of protein lysate were separated by SDS-PAGE, and boiled for 5 min. Subsequently, the samples were transferred to polyvinylidene difluoride (PVDF) membranes, and blocked with TTBS (0.5% Tween 20, 10 mM Tris-HCl, pH 7.5, 150 mM NaCl) containing 5% non-fat milk for 1 h at room temperature, and incubated with antibodies against NF-κB p-50 (5 μg/mL) and IκB-α (0.5 μg/mL) or GAPDH (0.2 μg/mL) overnight at 4 °C. The membranes were washed and further incubated with HRP-conjugated secondary antibodies against mouse NF-κB p-50 (1:5000) and IκB-α (1:5000) or GAPDH (1:5000) for 1 h at room temperature. After washing, the protein bands were detected using an ECL detection kit (Millipore, United States) and exposure to Kodak BioMax light films. The films were scanned into a computer and semi-quantified using NIH ImageJ software (Gao et al., 2018).

2.9. Immunohistochemical analyses

On day 70 of the experiments, we used immunohistochemical analysis methods to determine the positive cell expressions of NF-κB p-50, IκB-α and phospho- NF-κB p-50 in the cardiac tissues of mice. The finished TMA sections were placed in xylene and dewaxed, and then they were placed in an ethanol gradient (100%, 90%, 80%, and 70%, 1 min per concentration). We briefly rinsed the TMA sections with distilled water, and then rinsed them three times with PBS. Thereafter, the sections were placed in a beaker, to which we added 600 mL of ethylenediamine tetraacetic acid (EDTA) buffer. The beakers containing the TMA sections were placed in a pressure cooker for antigen repair for 10 min, after which we turned off the pressure cooker, and waited for the liquid in the beakers to cool naturally. After the liquid had cooled, we removed the sections and rinsed them once with distilled water, and three times with PBS. Subsequently, a 3% hydrogen peroxide aqueous solution was added, and sections were incubated at room temperature for 5 min to reduce non-specific background staining. We then placed the slices in the incubator box, added an anti-drip and incubated the slices at 4 °C overnight. The next day, slices were rinsed three times with

PBS. We added a drop of the En Vision reagent, incubated slices for 30 min at room temperature, and rinsed them with PBS three times. We subsequently added an appropriate amount of DAB chromatin to the slices, stained the slices for 2 to 10 min at room temperature whilst observing and controlling the staining time under a microscope, and stopped the coloring after the staining effect was brown or tan. Finally, we used hematoxylin to re-dye slices, washed slices in water, applied 1% ethanol differentiation for 30 s, washed slices in water again, placed slices in 1% ammonia water for 15 s, dehydrated the slices using an ethanol gradient, made the sections transparent using xylene, used neutral gum to seal the sections, and finally took photos of the slices. We quantitatively analyzed the images using Image-Pro Plus Version 6.0 software, using the following formula:

$$Positive\ cell\ expressions = \frac{IOD}{area} \times 100\%$$

Where IOD: integrated optical density; area: the area of the cells.

2.10. statistical analysis

Origin 8.0 software (MicroCal, United States) was used to perform the statistical analysis. Results are expressed as the mean ± standard error of the mean (SEM). All data were analyzed using one-way analysis of variance (ANOVA), followed by Scheffe's method. Differences with $P < 0.05$ were considered statistically significant.

3. Results

3.1. The blood lipid level of mice in each group on day 42

On day 42 of the experiments, the blood lipid levels of the mice were detected. Compared with CG, the levels of TC, TG and LDL-C in the serum of each model group were significantly increased, which was statistically significant ($P < 0.01$). The result is showing in Fig. 2.

3.2. Echocardiographic evaluation of effects of DLT on cardiac function in mice on day 70

Results of the echocardiographic evaluation of the effects of DLT on cardiac function in mice on day 70 of the experiment are shown in Fig. 3. The cardiac functions of mice are shown in Table 1. The left ventricular ejection fraction (LVEF, %) value, and left ventricular fractional shortening (LVFS, %) value were significantly lower in mice in HG, IG and HIG group as compared to the CG mice ($P < 0.01$). The LVEF value and LVFS value of the IG mice were significantly higher than those of the HIG mice ($P < 0.05$). After administration of DLT, the LVEF value and LVFS value increased in the HDG and HIDG mice ($P < 0.05$ and $P < 0.01$, respectively). Furthermore, the LVEF and LVFS values in the HDG mice were significantly higher than in the HIDG mice ($P < 0.01$).

The left ventricular masses (LVM) of mice are shown in Table 1. Compared with the CG mice, the LVM of mice in HG, IG and HIG groups were significantly higher ($P < 0.05$ or $P < 0.01$). The LVM of the IG mice were significantly higher than those of the HIG mice ($P < 0.05$). After administration of DLT, the LVM significantly decreased in the HDG mice, as compared to the HG mice ($P < 0.01$). The LVM of the HDG mice was significantly lower than those of the HIDG mice ($P < 0.01$).

The left ventricular volumes of mice are shown in Table 1. Compared with the CG mice, the left ventricular end-diastolic volumes (LV Vold) of the HG mice were significantly smaller, while the LV Vold of the IG mice was significantly larger ($P < 0.05$). The LV Vold of the HG mice were significantly smaller than those of the HIG mice ($P < 0.01$). After the administration of DLT, the LV Vold significantly decreased in the IDG mice, as compared to the IG mice ($P < 0.05$). Additionally, the LV Vold of the HDG mice and the IDG mice were significantly smaller than those of the HIDG mice ($P < 0.05$ or $P < 0.01$). Compared with the CG mice, the left ventricular end-systolic volumes (LV Vols) of HG, IG and HIG groups were significantly larger ($P < 0.05$ or $P < 0.01$). The LV Vols of the HG mice were significantly smaller than those of the HIG mice ($P < 0.05$). After the administration of DLT, the LV Vols significantly decreased in the HDG mice as compared to the HG mice ($P < 0.05$). Additionally, the LV Vols of the HDG mice were significantly smaller than those of the HIDG mice ($P < 0.01$).

The changes in the left ventricular structure in mice are shown in Table 1. Compared with the CG mice, the left ventricular internal dimensions at end-diastole (LVIDd) of the IG mice were significantly thicker ($P < 0.05$). The LVIDd of the IG mice were significantly thicker than those of the HIG mice ($P < 0.05$). Compared with the CG mice, the left ventricular posterior wall thickness at end-diastole (LVPWd) of the HG mice was significantly thicker ($P < 0.05$). After the administration of DLT, the LVIDd of the HDG mice were significantly smaller than those of the HIDG mice ($P < 0.05$), and the left ventricular internal dimensions at end-systole (LVIDs) of the HDG mice were significantly smaller than those of the HIDG mice ($P < 0.05$). The LVPWd of the IDG mice were significantly thicker than those of the IG mice ($P < 0.05$). The left ventricular posterior wall thicknesses at end-systole (LVPWs) of the HDG mice were significantly thicker than those of the HIDG mice ($P < 0.05$).

3.3. Histopathological examination of effects of DLT on cardiac tissue in mice

The histopathological examination of effects of DLT on cardiac tissue sections stained with H&E is shown in Fig. 4. In the CG mice, the aortic sinus was clear, and the inner membrane was smooth, and there was no plaque. Compared with the CG mice, the cardiac tissue showed obvious atherosclerotic lesions in the HG and HIG mice, while in the IG mice, the lesions were less obvious. In the HG and HIG mice, obvious thickening was observed under the inner membrane of the arteries. Valves were deformed and thickened. Valves and perivalvular tissues had obvious large areas consisting of necrotic nuclei, and a large number of cholesterol crystals appeared. After the administration of DLT, the histopathology of cardiac tissue in each treatment group was reversed to varying degrees.

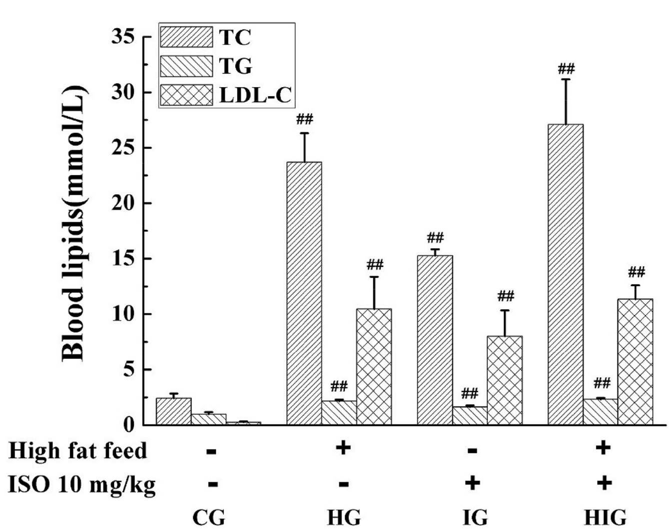

Fig. 2. Concentrations of the blood lipids of mice on day 42 of the experiment. CG: Control group; HG: High fat feed group; IG: ISO group; HIG: High fat feed + ISO group. Values are shown as means ± SEM ($n = 6$). $^{\#\#}P < 0.01$ each model group vs. CG.

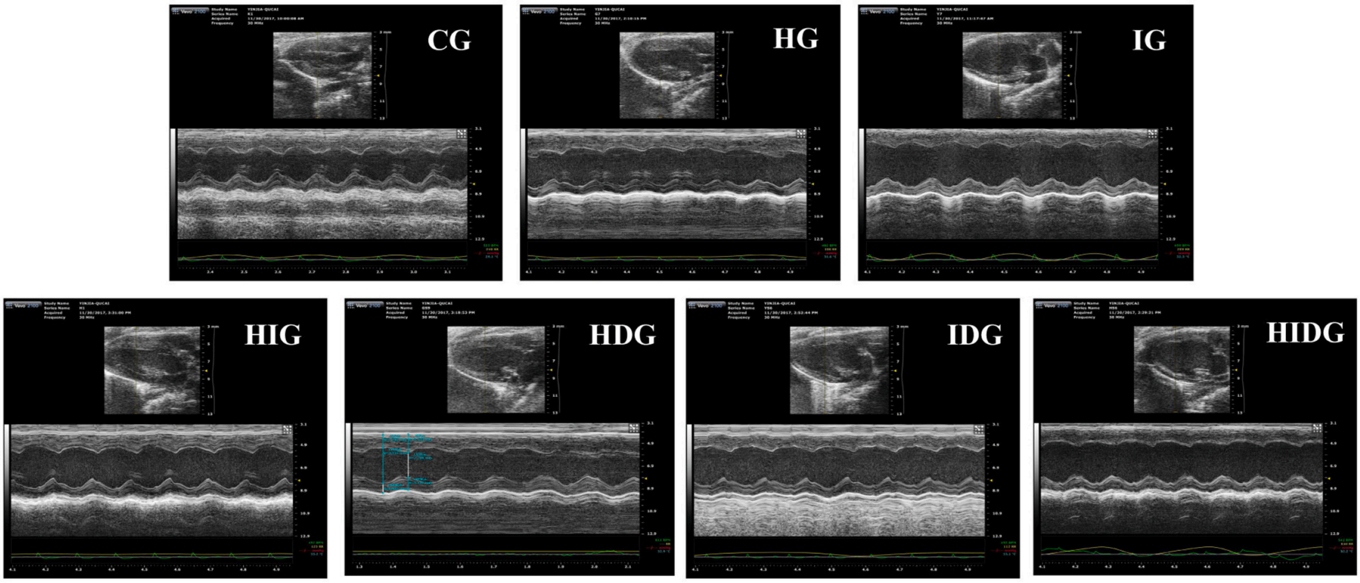

Fig. 3. Echocardiographic evaluations of the effects of DLT on cardiac function in mice on day 70 of the experiment. CG: Control group; HG: High fat feed group; IG: ISO group; HIG: High fat feed + ISO group; HDG: High fat feed + DLT group; IDG: ISO + DLT group; HIDG: High fat feed + ISO + DLT group.

3.4. Results of oil red O staining examination of effects of DLT on atherosclerotic plaques in the aortas of mice

Fig. 5 shows the results of oil red O staining examination of the effects of DLT on atherosclerotic plaques in the aortas of mice. CG mice had aortas that were smooth and flat, and there was no obvious red lipid plaque formation. Compared with the CG mice, there were massive lipid plaques in the aortas of HG and HIG mice, which were distributed throughout the aortas, and mainly concentrated in the aortic arch bifurcation and the lower abdominal aorta. In comparison, there were less red lipid plaques visible in the aorta of the IG mice. The ratios of aortic plaque area to vascular wall area of mice in each experimental group were significantly higher, as compared to the CG mice ($P < 0.01$), and the ratios of aortic plaque area to vascular wall area in the HG and IG mice were significantly smaller than those of the HIG mice ($P < 0.05$ or $P < 0.01$). DLT administration resulted in a lower total area of aortic plaques, and a significantly lower ratio of aortic plaque area to vascular wall area in mice in the HDG and HIDG groups ($P < 0.01$). Furthermore, the ratios of aortic plaque area to vascular wall area in the HDG mice were significantly smaller than those of the HIDG mice ($P < 0.05$).

3.5. Effects of DLT on concentrations of inflammatory factors

DLT administration resulted in lower concentrations of inflammatory factors associated with the NF-κB signaling pathway in the experimental mice to varying degrees. The ELISA tests showed that the concentrations of TNF-α in cardiac tissue of mice were significantly higher in the HG and HIG mice compared with the CG mice ($P < 0.05$ or $P < 0.01$, Fig. 6A). DLT treatment resulted in significantly lower concentrations of TNF-α in the IDG and HIDG mice ($P < 0.01$), and the concentrations of TNF-α in the IDG mice were significantly lower than those of the HIDG mice ($P < 0.01$).

Concentrations of IL-6 in cardiac tissue of mice were significantly higher in mice in each experimental group compared with the CG mice ($P < 0.05$ or $P < 0.01$, Fig. 6B). DLT administration resulted in significantly lower concentrations of IL-6 in the HIDG mice ($P < 0.05$), and the concentrations of IL-6 in the IDG mice were significantly higher than those in the HIDG mice ($P < 0.01$).

Concentrations of IL-1β in cardiac tissue of mice were significantly higher in the IG mice compared with the CG mice ($P < 0.01$, Fig. 6C), and the concentrations of IL-1β in the IG mice were significantly higher than those of the HIG mice ($P < 0.01$). DLT administration resulted in significantly lower concentrations of IL-1β in the HDG and IDG mice ($P < 0.01$), and the concentrations of IL-1β were significantly lower in the HDG mice than in the HIDG mice ($P < 0.05$). However, concentrations of IL-1β were significantly higher in IDG mice than in HIDG mice ($P < 0.05$).

Concentrations of serum IL-8 were significantly higher in the HG mice, and significantly lower in the IG mice, compared with the CG mice ($P < 0.01$, Fig. 6D). Concentrations of IL-8 in the HG mice were significantly higher than those of the HIG mice, but concentrations of IL-8 were significantly lower in IG mice than in HIG mice ($P < 0.05$). DLT administration resulted in significantly lower concentration of IL-8 in the HDG mice ($P < 0.01$), and the concentrations of IL-8 in the HDG and IDG mice were significantly lower than in the HIDG mice ($P < 0.05$ or $P < 0.01$).

3.6. Effects of DLT on the concentrations of MMP-1 and MMP-2

Concentrations of MMP-1 in the serum of mice are shown in Fig. 6E. The concentrations of MMP-1 were significantly higher in the HG and HIG mice compared with the CG mice ($P < 0.01$), and the concentrations of MMP-1 in the IG mice were significantly lower than those of the HIG mice ($P < 0.01$). DLT administration resulted in significantly lower concentrations of MMP-1 in the HDG and HIDG mice ($P < 0.05$ or $P < 0.01$).

Serum concentrations of MMP-2 are shown in Fig. 6F. Concentrations of MMP-2 were significantly higher in the HG and HIG mice compared with the CG mice ($P < 0.01$), and concentrations of MMP-2 in the IG mice were significantly lower than those of the HIG mice ($P < 0.01$). DLT administration resulted in significantly lower concentrations of MMP-2 in the HDG and HIDG mice ($P < 0.05$).

3.7. Effects of DLT on the protein expression of NF-κB p-50 and IκB-α

Protein expressions of NF-κB p-50 and IκB-α were detected by Western blotting, as shown in Fig. 7. Protein expression of NF-κB p-50 was significantly higher in the HG mice, and significantly lower in the IG and HIG mice, compared with the CG mice ($P < 0.01$, Fig. 7A and B), and the expression of NF-κB p-50 in the HG and IG mice was significantly higher than that of the HIG mice ($P < 0.01$). DLT administration resulted in a significantly lower expression of NF-κB p-50 in the HDG mice,

Table 1
Echocardiographic evaluation of DLT on cardiac function of mice on day 70 of the experiment.

Groups	*n*	LVEF (%)	LVFS (%)	LVM (mg)	LV Vold (μL)	LV Vols (μL)	heart rate	LVIDd (mm)	LVIDs (mm)	IVSd (mm)	IVSs (mm)	LVPWd (mm)	LVPWs (mm)
CG	15	59.91±1.18	31.34±0.85	90.76±1.60	54.93±1.07	23.42±0.93	520.33±4.25	3.65±0.06	2.61±0.06	1.17±0.06	1.16±0.07	0.77±0.02	1.02±0.03
HG	14	47.95±1.30$^{\#\#}$	23.56±0.75$^{\#\#}$	112.77±3.28$^{\#\#}$	50.85±1.05$^{\#\triangle\triangle}$	26.95±1.07$^{\#\triangle}$	506.50±5.63	3.46±0.06	2.63±0.06	1.16±0.05	1.59±0.03	0.86±0.03$^{\#}$	1.06±0.03
IG	13	49.93±1.33$^{\#\#\triangle}$	24.92±0.79$^{\#\#\triangle}$	121.87±5.92$^{\#\#\triangle}$	61.33±1.53$^{\#}$	29.05±1.13$^{\#\#}$	499.15±5.98	3.80±0.06$^{\#\triangle}$	2.80±0.07	1.19±0.07	1.64±0.06	0.76±0.02	0.97±0.05
HIG	14	44.02±1.15$^{\#\#}$	21.32±0.67$^{\#\#}$	109.50±2.20$^{\#}$	58.80±1.20	30.89±0.99$^{\#\#}$	508.79±5.20	3.64±0.05	2.76±0.06	1.07±0.03	1.54±0.03	0.83±0.01	0.97±0.03
HDG	14	60.17±0.78$^{**\blacktriangle\blacktriangle}$	31.36±0.56$^{**\blacktriangle\blacktriangle}$	88.93±2.67$^{**\blacktriangle\blacktriangle}$	49.19±1.63$^{\blacktriangle\blacktriangle}$	22.30±1.10$^{*\blacktriangle\blacktriangle}$	494.86±4.05	3.56±0.06$^{\blacktriangle}$	2.53±0.06$^{\blacktriangle}$	1.08±0.05	1.56±0.05	0.87±0.02	1.11±0.02$^{\blacktriangle}$
IDG	13	49.18±1.66	24.39±1.01	105.83±7.13	53.72±1.70$^{*\blacktriangle}$	26.38±1.23	497.62±7.81	3.62±0.06	2.76±0.07	1.18±0.08	1.64±0.07	0.84±0.03*	0.99±0.03
HIDG	14	51.62±1.77*	25.97±1.17*	104.46±2.72	59.14±1.17	27.98±1.31	506.29±4.54	3.77±0.06	2.89±0.08	1.13±0.03	1.57±0.04	0.84±0.04	0.98±0.05

CG: Control group; HG: High fat feed group; IG: ISO group; HIG: High fat feed + ISO group; HDG: High fat feed + DLT group; IDG: ISO + DLT group; HIDG: High fat feed + ISO + DLT group. Values are shown as means ± SEM. $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ each model group vs. CG. $^{*}P < 0.05$, $^{**}P < 0.01$ each treatment group vs. its corresponding model group. $^{\triangle}P < 0.05$, $^{\triangle\triangle}P < 0.01$, HG or IG vs. HIG. $^{\blacktriangle}P < 0.05$, $^{\blacktriangle\blacktriangle}P < 0.01$, HDG or IDG vs. HIDG.

compared with that of the HG mice ($P < 0.01$). DLT administration resulted in a significantly lower expression of NF-κB p-50 in the IDG mice compared with that of the IG mice ($P < 0.05$). Additionally, the expression of NF-κB p-50 in the HDG mice was significantly higher than that of the HIDG mice ($P < 0.05$).

Protein expression of IκB-α was significantly higher in the HG mice, and was significantly lower in the IG and HIG mice, compared with the CG mice ($P < 0.05$ or $P < 0.01$, Fig. 7A and C), and the expression of IκB-α was significantly higher in the HG mice, but expression of IκB-α in the IG mice was significantly lower than that of the HIG mice ($P < 0.01$). DLT administration resulted in significantly lower expression of IκB-α in the HDG mice compared with that of the HG mice, but DLT administration significantly increased the expression of IκB-α in the IDG mice compared with that of the IG mice ($P < 0.01$). Additionally, the expression of IκB-α in the IDG mice was significantly lower than that in the HIDG mice ($P < 0.05$).

3.8. *Effects of DLT on the positive cell expressions of NF-κB p-50, IκB-α and phospho-NF-κB p-50*

Results of the immunohistochemical analyses are shown in Fig. 8A, C and 8E. The positive cell expressions of NF-κB p-50 were significantly higher in mice in HG, IG and HIG group compared with the CG mice ($P < 0.05$ or $P < 0.01$, Fig. 8B), and the positive cell expressions of NF-κB p-50 in the HG and IG mice were significantly lower than those of the HIG mice ($P < 0.05$ or $P < 0.01$). DLT administration resulted in significantly lower positive cell expressions of NF-κB p-50 in HDG, IDG and HIDG groups ($P < 0.05$ or $P < 0.01$), and the positive cell expressions of NF-κB p-50 in the HDG and IDG mice were significantly lower than those of the HIDG mice ($P < 0.01$).

The positive cell expressions of IκB-α were significantly higher in mice in each experimental group compared with the CG mice ($P < 0.05$ or $P < 0.01$, Fig. 8D), and the positive cell expressions of IκB-α in the IG mice were significantly lower than that of the HIG mice ($P < 0.01$). DLT administration resulted in significantly lower positive cell expressions of IκB-α in the HDG and HIDG mice ($P < 0.05$ or $P < 0.01$), and the positive cell expressions of IκB-α in the HDG and IDG mice were significantly lower than that in the HIDG mice ($P < 0.05$).

The positive cell expressions of phospho-NF-κB p-50 were significantly higher in HG, IG and HIG group compared with the CG mice($P < 0.01$, Fig. 8F), and the positive cell expressions of phospho-NF-κB p-50 in HG and IG mice was significantly lower than those of the HIG mice ($P < 0.05$ or $P < 0.01$). DLT administration resulted in significantly lower positive cell expressions of phospho-NF-κB p-50 in HDG and HIDG groups ($P < 0.01$), and the positive cell expressions of phospho-NF-κB p-50 in the IDG mice were significantly higher than that in the HIDG mice ($P < 0.05$).

4. Discussion

ApoE is very important for the normal catabolism of triglyceride proteins. A lack of ApoE can lead to the accumulation of cholesterol in the circulating blood, and is likely to cause the formation of atherosclerosis lesions (Wang et al., 2017). ApoE$^{-/-}$ mice can develop atherosclerosis when fed a natural diet, and the pathological changes are very similar to those in humans. ApoE$^{-/-}$ mice can even form fibrous plaques, and, as such, they are widely used in atherosclerosis research (Auclair et al., 2009; Hauser et al., 2011; Chen et al., 2018). Therefore, the ApoE$^{-/-}$ mouse model is considered to be an ideal model for studies on atherosclerosis interventions. Some studies suggest that, compared with the ApoE$^{-/-}$ mice fed normal diets, the atherosclerotic plaques and the number of inflammatory cells significantly increase, the matrix fiber components in the fiber caps decrease, the shape of the fiber cap thins, and the deposit of neutral lipid droplets increases in ApoE$^{-/-}$ mice fed a high-fat diet (Zhou et al., 2009). The ApoE$^{-/-}$ mouse model is relatively stable, and the molding time is relatively short. Though the cost of

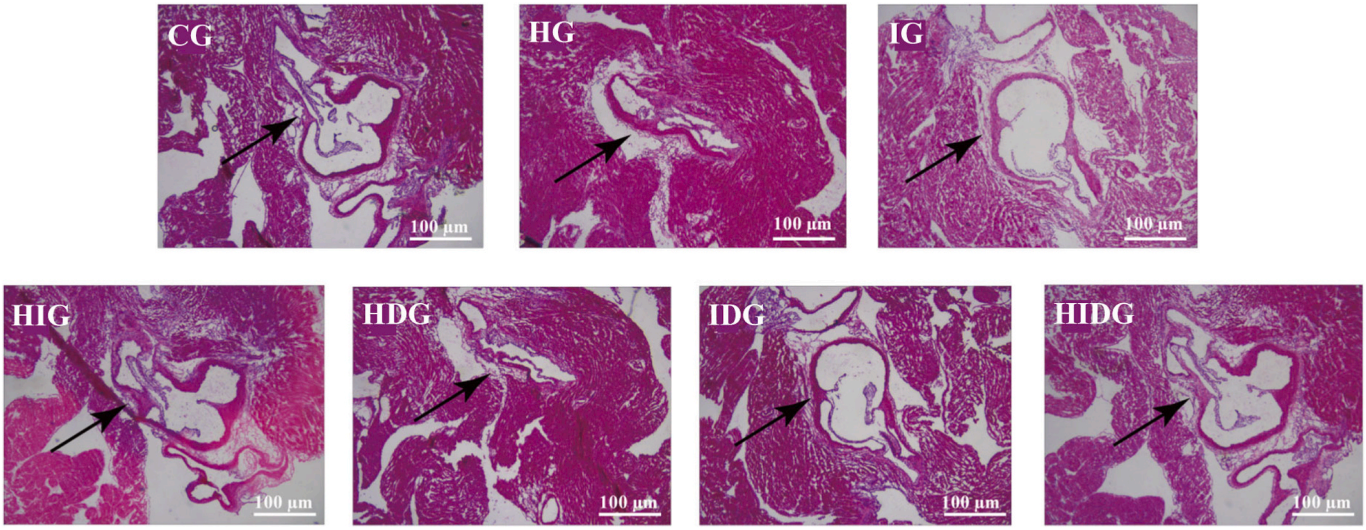

Fig. 4. Histopathological examination of DLT on cardiac tissue sections stained with H&E (100 × magnification). CG: Control group; HG: High fat feed group; IG: ISO group; HIG: High fat feed + ISO group; HDG: High fat feed + DLT group; IDG: ISO + DLT group; HIDG: High fat feed + ISO + DLT group.

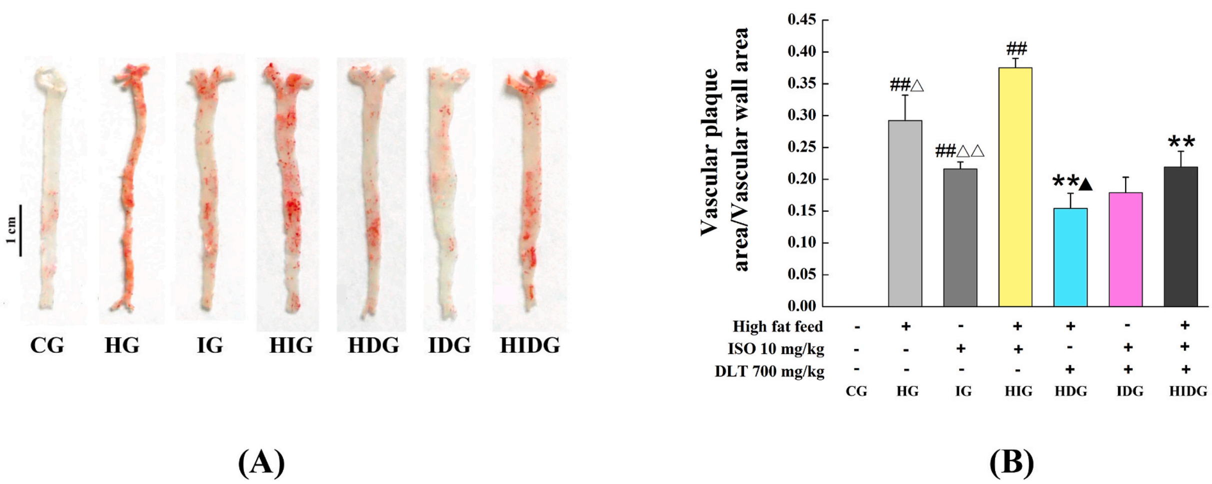

Fig. 5. (A) Oil red O staining of DLT on atherosclerotic plaques in the aortas of mice. CG: Control group; HG: High fat feed group; IG: ISO group; HIG: High fat feed + ISO group; HDG: High fat feed + DLT group; IDG: ISO + DLT group; HIDG: High fat feed + ISO + DLT group. **(B)** The ratio of vascular plaque area to vascular wall area of the aorta. Values are shown as means ± SEM (n = 3). $^{\#\#}P < 0.01$ each model group vs. CG. $^{**}P < 0.01$ each treatment group vs. its corresponding model group. $^{\triangle}P < 0.05$, $^{\triangle\triangle}P < 0.01$, HG or IG vs. HIG. $^{\blacktriangle}P < 0.05$, HDG vs. HIDG. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

genetically modified animal models is relatively high, ApoE$^{-/-}$ mice provide a good model to study the process of human disease occurrence and development (Chen et al., 2001; Zadelaar et al., 2007). According to the references, we chose the six-week ApoE$^{-/-}$ mice in order to avoid the occurrence of combined diseases and weak resistance after the ApoE$^{-/-}$ mice fed a high fat diet combined with isoprenaline injection, and reduce the probability of other uncertainties (Aoyama et al., 2009; Shimada et al., 2011). Therefore, the ApoE$^{-/-}$ mouse model was used in our study in order to investigate the mechanism of the effects of DLT on AWMI.

The myocardial ischemia model created by subcutaneous injection with ISO is a relatively non-invasive and simple classical model, and the technology is established and widely used. This method can replicate more stable myocardial ischemia and fibrosis models, and it also has obvious advantages in drug pre-clinical research evaluations and drug screening. Most of the experiments used ligation of the left anterior descending coronary artery or large dose of ISO injected in the animal to prepare heart failure model (Garson et al., 2015; Tang et al., 2015; Chang et al., 2018). In this study, the animal model of AWMI was established using ApoE$^{-/-}$ mice fed a high fat diet combined with a small dose and multiple ISO injections in order to observe the process of myocardial ischemia to heart failure and compare the different effects of DLT among the three treatment groups. The mechanism for establishment of AWMI model is that the high-viscosity blood flow originally caused by high-fat became more blocked, forming the microcirculation disorder, so that the ApoE knockout mice had atherosclerosis and combined with the state of myocardial ischemia, thus establishing the AWMI model. As such, it is easier to create a hyperlipidemia state in the ApoE$^{-/-}$ mice fed a high-fat diet than in ordinary mice (Changes in body weight in each group of mice per week are listed in Supplementary Table S1. Concentrations of the blood lipids of mice on day 42 of experiments are listed in Fig. 2), which can shorten the modeling time, especially if ApoE$^{-/-}$ mice are simultaneously provided with a small dose of ISO. The results of blood lipids showed that compared with CG, the levels of blood lipids (TC, TG and LDL-C) in each model group increased to varying degrees. The results of echocardiographic measurement showed that the changes in LVM and left ventricular volume occurred in the early stages of the experiment in the model mice, followed by a

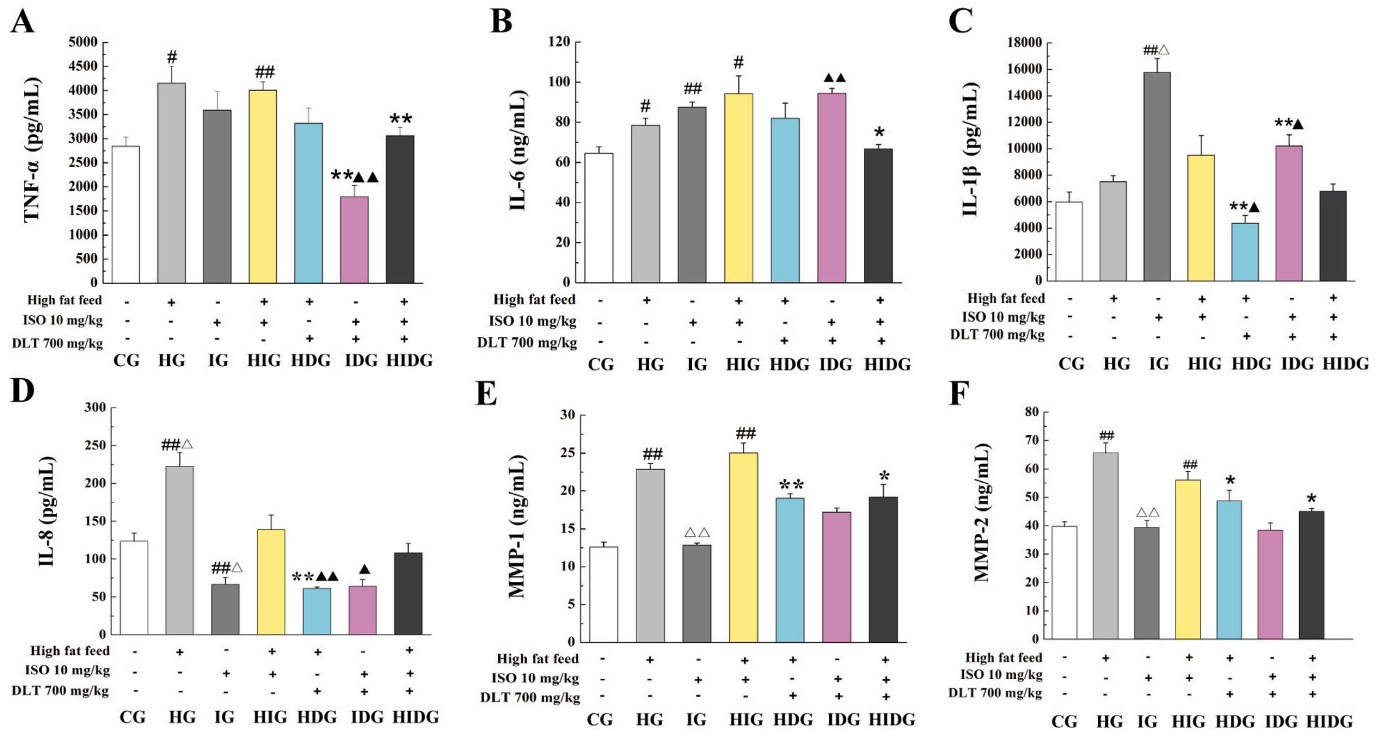

Fig. 6. Effects of DLT on the concentrations of inflammatory factors and MMP in mice. **(A)** TNF-α, **(B)** IL-6 and **(C)** IL-1β concentrations in the cardiac tissue; **(D)** IL-8, **(E)** MMP-1 and **(F)** MMP-2 concentrations in the plasma. CG: Control group; HG: High fat feed group; IG: ISO group; HIG: High fat feed + ISO group; HDG: High fat feed + DLT group; IDG: ISO + DLT group; HIDG: High fat feed + ISO + DLT group. Values are shown as means ± SEM ($n = 5$). $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ each model group vs. CG. $^{*}P < 0.05$, $^{**}P < 0.01$ each treatment group vs. its corresponding model group. $^{\triangle}P < 0.05$, $^{\triangle\triangle}P < 0.01$, HG or IG vs. HIG. $^{\blacktriangle}P < 0.05$, $^{\blacktriangle\blacktriangle}P < 0.01$, HDG or IDG vs. HIDG.

significant decrease in LVEF and LVFS values, cardiac hypertrophy with enlarged cardiac cavity, and the myocardial ischemia state (the echocardiographic evaluations in mice on day 42 of the experiment are shown in Supplementary Fig. S1; Tables S2 and S3). Ultrasonic examination found that compared with CG, the left ventricular LVEF and LVFS values of each model group were significantly reduced, LVM was significantly increased, LVPWd was significantly thinner in the IG, and LVPWs in the HG and IG were significantly thinner ($P < 0.05$ or $P < 0.01$); After intervention of DLT, the results showed that DLT can significantly increase the left ventricular LVEF and LVFS of HDG and HIDG mice compared with their corresponding model groups, reduce LVM of mice in each treatment groups, reduce LV Vold in IDG and LV Vols in HDG ($P < 0.05$ or $P < 0.01$). These results show that DLT can improve the left ventricular structure and morphology of AWMI mice to different degrees. Therefore, we used the ApoE$^{-/-}$ mice fed a high fat feed combined with ISO injection to establish the atherosclerosis mice model with myocardial ischemia. The results of detecting the cardiac function, blood lipid level, histopathology of cardiac tissue, and atherosclerotic plaques in the aortas showed that this model could simulate the state of AWMI. After the DLT administration, the left ventricular structure and morphology of the mice, histopathology of cardiac tissue, and the area of aortic plaques all improved to varying degrees, and there were certain differences among the three treatment groups.

The purpose of this study is to observe the difference between the same model before and after DLT administration, that is, to compare the HDG with HG, IDG with IG, HIDG with HIG. In order to the integrity of the study data, on the other hand, this study conducted a comparison between different model groups. Although there are differences between the HG, IG and HIG models, we hope to observe the effects of same drugs on the three model groups of HG, IG, and HIG through quantitative or qualitative experiments. That is, we used the same drug to treat different model groups in order to observe what results and regular findings will occur. It suggested that DLT has different effects on different models, but for the same detection index, sometimes similar effects appear, showing the same trend.

The inflammation hypothesis of atherosclerosis suggested that atherosclerosis is a chronic inflammatory disease. Pathological studies and epidemiological investigations show that inflammation plays an important role in the occurrence and development of atherosclerosis, and there are varying degrees of local and systemic inflammatory reactions (Libby et al., 2010; Lyu et al., 2018). A variety of inflammatory factors (such as TNF-α, IL-1β, IL-6, IL-8) could accelerate the development of atherosclerosis, leading to deterioration of the condition, and adverse cardiovascular events (Newby, 2005). The activation of NF-κB, a proinflammatory transcription factor, is a commonly observed phenomenon in atherosclerosis, hypertension, diabetes, cancer and so on (Wang et al., 2015a,b,c; Qin et al., 2017; Zhang et al., 2017; Bhat et al., 2018; Oyagbemi et al., 2018). NF-κB is an important inflammatory transcription factor which is a dimer composed of polypeptide chains p-65 and p-50 protein subunits, and its combined forms include p-65 homodimer, p-50 homodimer polymers and p-65-p-50 heterodimers and belongs to the Rel protein family. At rest, the heterodimer of NF-κB (p-65-p-50) binds to the NF-κB inhibitor protein (IκB) and exists in the cytoplasm in an inactive form. When NF-κB is blocked by TNF-α, IL-1 and other cytokines, it promotes the phosphorylation of N-terminal serine of IκB, which in turn activates NF-κB. Then it translocates into the nucleus, and regulates the expression of various inflammatory factors, adhesion molecules and metalloproteinases (MMPs) at the transcription level. Therefore, NF-κB is considered to be one of the key targets in the initial mechanism of vascular endothelial cell damage. These inflammatory cytokines are closely related to the occurrence of a series of lesions such as lipid deposition in atherosclerosis, migration and proliferation of smooth muscle cells, aggregation of monocytes, foam cell formation, and apoptosis, which in turn promote the development of atherosclerosis and affect the stability of plaques (Weber and Noels, 2011; Hwang et al., 2013). These inflammatory factors, in turn, can

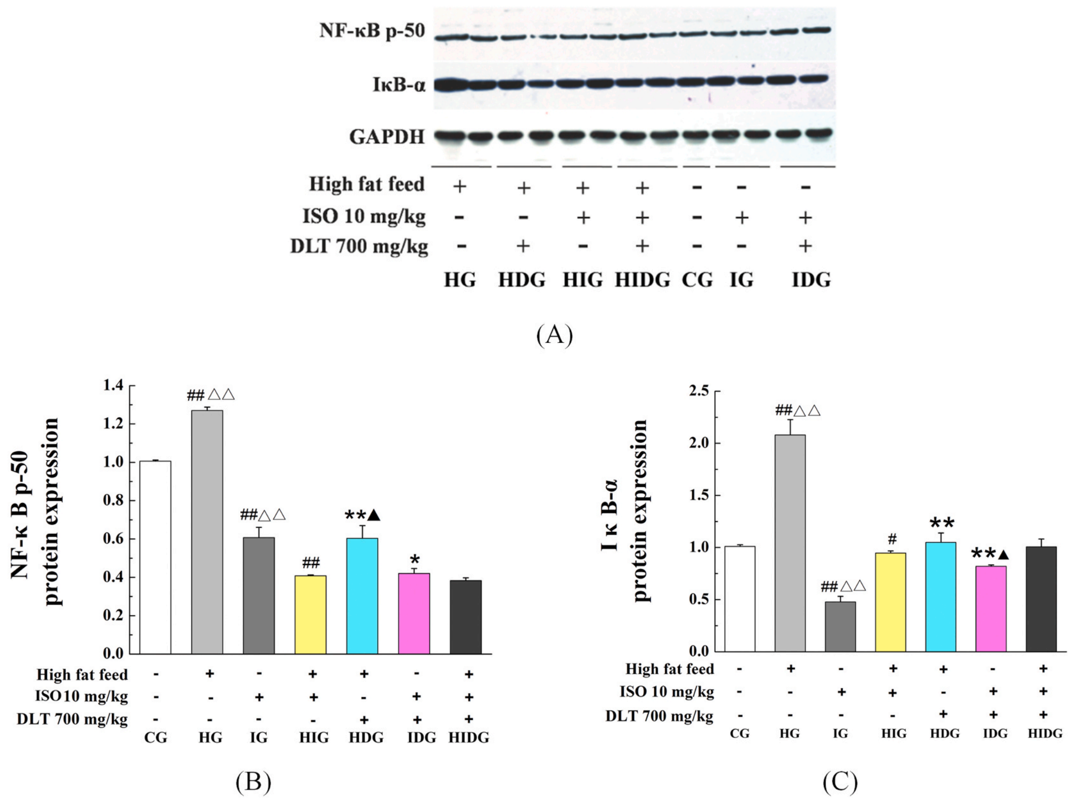

Fig. 7. Effects of DLT on the protein expression of NF-κB p-50 **(A, B)** and IκB-α **(A, C)** in cardiac tissues of mice. CG: Control group; HG: High fat feed group; IG: ISO group; HIG: High fat feed + ISO group; HDG: High fat feed + DLT group; IDG: ISO + DLT group; HIDG: High fat feed + ISO + DLT group. Values are shown as means ± SEM ($n = 3$). $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ each model group vs. CG. $^{*}P < 0.05$, $^{**}P < 0.01$ each treatment group vs. its corresponding model group. $^{\triangle\triangle}P < 0.01$, HG or IG vs. HIG. $^{\blacktriangle}P < 0.05$, HDG or IDG vs. HIDG.

inhibit the activation of NF-κB, forming an inflammatory cascade reaction that accelerates the occurrence of atherosclerosis. Therefore, NF-κB is considered to be one of the key targets in the mechanism of vascular endothelial cell damage. As such, we proposed that the NF-κB signaling pathway regulated cholesterol metabolism AWMI should be investigated *in vivo*.

In this study, after the administration of DLT, the results showed that the concentrations of TNF-α, IL-6, IL-1β and IL-8 of mice decreased to varying degrees. In the Western Blot study, the results showed that the protein expression of NF-κB p-50 decreased significantly in the HDG and IDG mice ($P < 0.05$ or $P < 0.01$), and the protein expression of IκB-α decreased significantly in the HDG mice ($P < 0.01$). The immunohistochemical analysis showed that the expression levels of NF-κB p-50, IκB-α and phospho-NF-κB p-50 positive cells in each model group increased to varying degrees, and there were certain differences between different syndrome types. DLT decreased the expression of the positive cells of NF-κB p-50 in various treatment groups to varying degrees ($P < 0.05$ or $P < 0.01$), and the positive cell expressions of IκB-α decreased significantly in HDG and HIDG mice ($P < 0.05$ or $P < 0.01$), and the positive cell expressions of phospho-NF-κB p-50 decreased significantly in HDG and HIDG mice ($P < 0.01$). The results suggest that DLT could play a therapeutic role by regulating NF-κB signaling pathway related inflammatory factors, including TNF-α, IL-6, IL-1β, IL-8, the protein expressions, and the positive cell expressions of NF-κB p-50, IκB-α and phospho-NF-κB p-50 in the model mice. We found different effects of DLT on the HDG, IDG, and HIDG mice. All three model groups use ApoE$^{-/-}$mice, thus mice themselves have certain obstacles in lipid metabolism. Therefore, the HG, IG and HIG all have a tendency to increase blood lipid levels. The HG and HIG mice were fed with high-fat diet, so the blood lipid levels of mice in these two groups were more obvious than that of IG alone, and there were statistical differences. The IG and HIG mice had obvious similar trends in the changes of the cardiac function. We suggest that the reason for the different effects of DLT on the HDG, IDG, and HIDG mice might be related to the fact that HG, IG and HIG groups have different pathologic mechanisms. Taken together, the analysis showed that the relevant targeting molecules mainly included TNF-α, IL-6, IL-1β, IL-8, NF-κB p-50, IκB-α and phospho-NF-κB p-50.

In addition, MMPs are the main proteases that degrade the extracellular matrix. They also play a role in the entire process of occurrence, development, and plaque rupture of atherogenesis (Wang et al., 2015a, b,c; Gresele et al., 2017). The formation and stability of plaque is related to the metabolism of the extracellular matrix. The direct cause of extracellular matrix degradation is the increase of MMPs expression (Tyagi et al., 1996; Fujimoto et al., 2008). A previous study found that the levels of MMP-1 expression in the serum of patients with unstable angina pectoris of coronary heart disease increased, suggesting that this may be related to the pathogenesis of unstable atherosclerotic plaques (Wang et al., 2013). In this study, the concentrations of MMP-1 and MMP-2 were significantly higher in the HG, IG and HIG groups. After the DLT administration, there were certain differences among the HDG, IDG and HIDG groups. The intervention effect of DLT on the concentrations of MMP-1 and MMP-2 in the HDG and HIDG mice was more obvious and had statistical significance. This indicates that the inhibitory effect of DLT on MMP-1 and MMP-2 might be one of the mechanisms by which it prevents AWMI.

5. Conclusion

In this study, we found that the administration of DLT is effective, although to a varying degree, for the treatment of atherosclerosis, myocardial ischemia, or AWMI in mice. The therapeutic efficacy of DLT

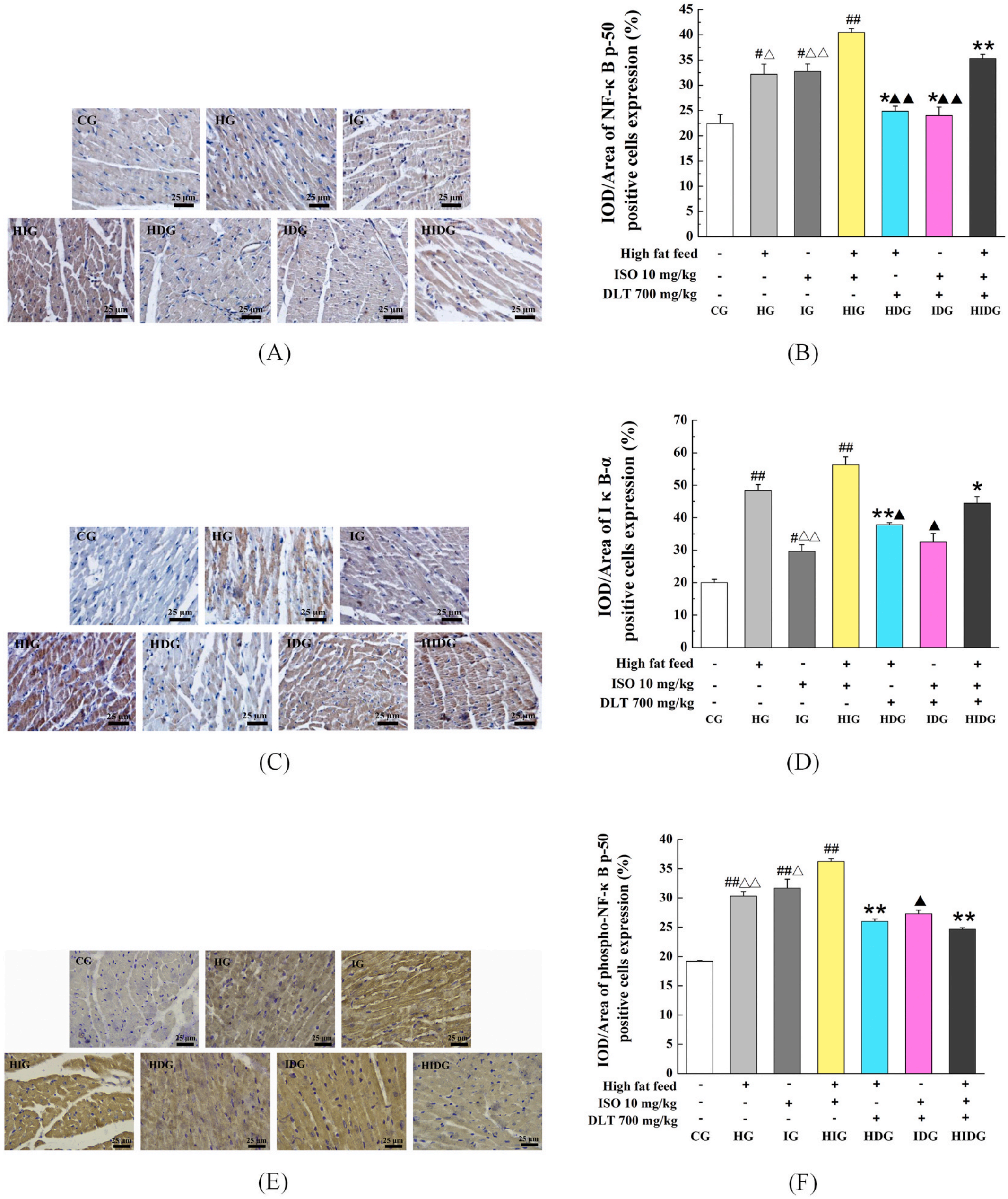

Fig. 8. Effects of DLT on the immunohistochemical analysis of NF-κB p-50 **(A)**, IκB-α **(C)** and phospho-NF-κB p-50 **(E)** of mice (400 × magnification). The positive cell expressions of NF-κB p-50 **(B)**, IκB-α **(D)** and phospho-NF-κB p-50 **(F)**. CG: Control group; HG: High fat feed group; IG: ISO group; HIG: High fat feed + ISO group; HDG: High fat feed + DLT group; IDG: ISO + DLT group; HIDG: High fat feed + ISO + DLT group. Values are shown as means ± SEM ($n = 3$). $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ each model group vs. CG. $^{*}P < 0.05$, $^{**}P < 0.01$ each treatment group vs. its corresponding model group. $^{\triangle}P < 0.05$, $^{\triangle\triangle}P < 0.01$, HG or IG vs. HIG. $^{\blacktriangle}P < 0.05$, $^{\blacktriangle\blacktriangle}P < 0.01$, HDG or IDG vs. HIDG.

in the HG, IG and HIG groups could be exerted at least in part by regulating the NF-κB signaling pathway. In future, we plan to study and further evaluate the mechanism of treating AWMI with DLT. In addition, the relationship between the quantity and effect of the treatment of AWMI by DLT still needs further investigation.

Author contributions

YZ and C-QY conceived and designed the study. ZL, JY, SW, RL contributed to experiments performance and data collection. LL, S-MG, and Y-LX participated in data evaluation. SG and X-XX drafted and reviewed the manuscript. L-NG provided guidance to the study. All authors read and approved the final manuscript.

Declaration of competing interest

All the authors declare that they have no conflict of interest. We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work. There is no professional or other personal interest of any nature or kind in any product, service and/or company that could be interpreted as influencing the work described in this manuscript.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 81403198) and National Key Basic Research Program (973 project, No. 2014CB542902). The authors would like to thank Jilin

Connell Medicine Co., Ltd. for providing DLT.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jep.2020.113158.

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